

Multivariate Optimization Formulations for DriftXpress Algorithms

Common notation

Let $\mathcal{D} = \{(x_i, c_i)\}_{i=1}^N$, with $x_i \in [0, 1]^D$, $D = 784$, and $c_i \in \{1, \dots, C\}$, $C = 6$. For $\xi_i \sim \mathcal{N}(0, I_d)$,

$$z_i^+ = \mu_\phi(x_i) + \exp(\tfrac{1}{2}\ell_\phi(x_i)) \odot \xi_i, \quad z_i^- = G_\theta(\varepsilon_i, c_i), \quad \hat{x}_i = D_\psi(z_i^+). \quad (1)$$

For temperature $T > 0$, define $k_T(u, v) = \exp(-\|u - v\|_2/T)$. For each class c , let $\mathcal{S}_c$ be the fixed positive-latent support, U_c its landmarks, $W_c = K_{U_c U_c}^{-1/2}$, and

$$\varphi_c(z) = W_c^\top k_T(U_c, z), \quad A_c = \sum_{u \in \mathcal{S}_c} \varphi_c(u) u^\top, \quad b_c = \sum_{u \in \mathcal{S}_c} \varphi_c(u). \quad (2)$$

The class-conditional attraction and batch repulsion fields are

$$m_c^+(z) = \frac{A_c^\top \varphi_c(z)}{\varphi_c(z)^\top b_c + \epsilon}, \quad (3)$$

$$m^-(z_b; Z) = \frac{\sum_{j \neq b} k_T(z_b, z_j) z_j}{\sum_{j \neq b} k_T(z_b, z_j) + \epsilon}, \quad (4)$$

$$V_b(Z, c) = m_{c_b}^+(z_b) - m^-(z_b; Z). \quad (5)$$

For a batch $Z = (z_b)_{b=1}^B$, define

$$\mathcal{D}(Z, c) = \frac{1}{Bd} \sum_{b=1}^B \|z_b - \text{sg}(z_b + V_b(Z, c))\|_2^2, \quad (6)$$

$$\mathcal{V}(Z) = \frac{1}{d} \sum_{j=1}^d \left[\gamma - \sqrt{\text{Var}_b(Z_{bj}) + \epsilon_v} \right]_+, \quad (7)$$

$$\mathcal{C}(Z) = \frac{1}{d} \left\| \text{offdiag} \left(\frac{\bar{Z}^\top \bar{Z}}{B-1} \right) \right\|_F^2, \quad (8)$$

$$\mathcal{R}(X, \hat{X}) = \frac{1}{BD} \|X - \hat{X}\|_F^2, \quad (9)$$

$$\mathcal{K}(\mu, \ell) = \frac{1}{2B} \sum_{b=1}^B \sum_{j=1}^d (\mu_{bj}^2 + e^{\ell_{bj}} - 1 - \ell_{bj}). \quad (10)$$

Algorithm 1: joint conditional training

$$\min_{\phi, \psi, \theta} \mathbb{E} \left[\mathcal{R}(X, D_\psi(Z^+)) + \lambda_{\text{KL}} \mathcal{K}(\mu_\phi, \ell_\phi) + \lambda_d^{(t)} \mathcal{D}(Z^-, c) + \lambda_v (\mathcal{V}(Z^+) + \mathcal{V}(Z^-)) + \lambda_c (\mathcal{C}(Z^+) + \mathcal{C}(Z^-)) \right] \quad (11)$$

where

$$\lambda_d^{(t)} = \lambda_d \min \left\{ 1, 1 + 3 \max \left(0, 1 - \frac{t}{30} \right) \right\}.$$

With the cache fixed, the objective is computationally separable:

$$\mathcal{L}_1 = \underbrace{\mathcal{L}_1^+(\phi, \psi)}_{\mathcal{R} + \lambda_{\text{KL}} \mathcal{K} + \lambda_v \mathcal{V}(Z^+) + \lambda_c \mathcal{C}(Z^+)} + \underbrace{\mathcal{L}_1^-(\theta)}_{\lambda_d^{(t)} \mathcal{D}(Z^-, c) + \lambda_v \mathcal{V}(Z^-) + \lambda_c \mathcal{C}(Z^-)}. \quad (12)$$

Algorithm 1 uses one loop and one optimizer object, but it is not fully joint in the computational-graph sense: $\mathcal{L}_1^+$ updates (ϕ, ψ) and $\mathcal{L}_1^-$ updates only θ because $Z^- = G_\theta(\varepsilon, c)$ bypasses D_ψ and μ_ϕ .

Algorithm 1 Algorithm 1 training

Require: $\mathcal{D}$, T , landmark count r , step size η , weights λ

Ensure: (ϕ, ψ, θ)

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1: Initialize  $(\phi, \psi, \theta)$ ; build fixed  $\{U_c, W_c, A_c, b_c\}_{c=1}^C$ 
2: for  $t = 0, \dots, T_{\text{train}}$  do
3:   Sample minibatch  $(X, c)$  and noise  $(\Xi, \varepsilon)$ 
4:    $Z^+ \leftarrow q_\phi(X; \Xi)$ ;  $\hat{X} \leftarrow D_\psi(Z^+)$ ;  $Z^- \leftarrow G_\theta(\varepsilon, c)$ 
5:   Compute  $V(Z^-, c)$  and  $\mathcal{L}_1^+, \mathcal{L}_1^-$  from (12)
6:    $(\phi, \psi) \leftarrow (\phi, \psi) - \eta \nabla_{\phi, \psi} \mathcal{L}_1^+$ ;  $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_1^-$ 
7: end for
8: return  $(\phi, \psi, \theta)$ 

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Algorithm 2: separate VAE and generator optimization

Stage I (VAE):

$$(\hat{\phi}, \hat{\psi}) \in \arg \min_{\phi, \psi} \mathbb{E}[\mathcal{R}(X, D_\psi(Z^+)) + \lambda_{\text{KL}} \mathcal{K}(\mu_\phi, \ell_\phi) + \lambda_v \mathcal{V}(\mu_\phi(X)) + \lambda_c \mathcal{C}(\mu_\phi(X))]. \quad (13)$$

Stage II (conditional generator):

$$\boxed{\hat{\theta} \in \arg \min_{\theta} \mathbb{E}[\lambda_d \mathcal{D}(G_\theta(\varepsilon, c), c) + \lambda_v \mathcal{V}(G_\theta(\varepsilon, c)) + \lambda_c \mathcal{C}(G_\theta(\varepsilon, c))]} \quad \text{subject to } (\phi, \psi) = (\hat{\phi}, \hat{\psi}) \text{ fixed.} \quad (14)$$

The VAE is optimized first and then frozen; the second optimization updates only the conditional generator and contains no reconstruction or KL term.

Algorithm 2 Algorithm 2 training

Require: $\mathcal{D}$, T , landmark count r , step size η , weights λ

Ensure: $(\hat{\phi}, \hat{\psi}, \theta)$

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1: Initialize  $(\phi, \psi)$ 
2: for  $t = 0, \dots, T_{\text{VAE}}$  do
3:   Sample  $(X, c)$  and  $\Xi$ ; form  $Z^+ = q_\phi(X; \Xi)$  and  $\hat{X} = D_\psi(Z^+)$ 
4:    $\mathcal{L}_{\text{VAE}} \leftarrow$  objective (13)
5:    $(\phi, \psi) \leftarrow (\phi, \psi) - \eta \nabla_{\phi, \psi} \mathcal{L}_{\text{VAE}}$ 
6: end for
7: Freeze  $(\hat{\phi}, \hat{\psi})$ ; build fixed  $\{U_c, W_c, A_c, b_c\}_{c=1}^C$ 
8: Initialize  $\theta$ 
9: for  $t = 0, \dots, T_G$  do
10:  Sample  $(c, \varepsilon)$ ; set  $Z^- \leftarrow G_\theta(\varepsilon, c)$ 
11:   $\mathcal{L}_G \leftarrow$  objective (14)
12:   $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_G$ 
13: end for
14: return  $(\hat{\phi}, \hat{\psi}, \theta)$ 

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Algorithm 3: fully joint training with re-encoded drift

Define

$$z_i^- = G_\theta(\varepsilon_i, c_i), \quad x_i^- = D_\psi(z_i^-), \quad \tilde{z}_i^- = \mu_\phi(x_i^-). \quad (15)$$

The optimization problem is

$$\boxed{\min_{\phi, \psi, \theta} \mathbb{E}[\mathcal{R}(X, D_\psi(Z^+)) + \lambda_{\text{KL}} \mathcal{K}(\mu_\phi, \ell_\phi) + \lambda_d \mathcal{D}(\tilde{Z}^-, c) + \lambda_v (\mathcal{V}(\mu_\phi(X)) + \mathcal{V}(\tilde{Z}^-)) + \lambda_c (\mathcal{C}(\mu_\phi(X)) + \mathcal{C}(\tilde{Z}^-))]} \quad (16)$$

Algorithm 3 is fully joint: drift is evaluated after decode-re-encode, so its gradient traverses $\tilde{Z}^- \leftarrow D_\psi \leftarrow G_\theta$ and $\tilde{Z}^- \leftarrow \mu_\phi$. No additional cycle loss is used.

Algorithm 3 Algorithm 3 training

Require: $\mathcal{D}$, T , landmark count r , step size η , weights λ

Ensure: (ϕ, ψ, θ)

- 1: Initialize (ϕ, ψ, θ) ; build fixed $\{U_c, W_c, A_c, b_c\}_{c=1}^C$
 - 2: **for** $t = 0, \dots, T_{\text{train}}$ **do**
 - 3: Sample (X, c) and (Ξ, ε)
 - 4: $Z^+ \leftarrow q_\phi(X; \Xi)$; $\hat{X} \leftarrow D_\psi(Z^+)$
 - 5: $Z^- \leftarrow G_\theta(\varepsilon, c)$; $X^- \leftarrow D_\psi(Z^-)$; $\tilde{Z}^- \leftarrow \mu_\phi(X^-)$
 - 6: Compute $V(\text{sg}(\tilde{Z}^-), c)$ and $\mathcal{L}_t$ from (16)
 - 7: $(\phi, \psi, \theta) \leftarrow (\phi, \psi, \theta) - \eta \nabla_{\phi, \psi, \theta} \mathcal{L}_t$
 - 8: **end for**
 - 9: **return** (ϕ, ψ, θ)
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Loss-to-parameter flow

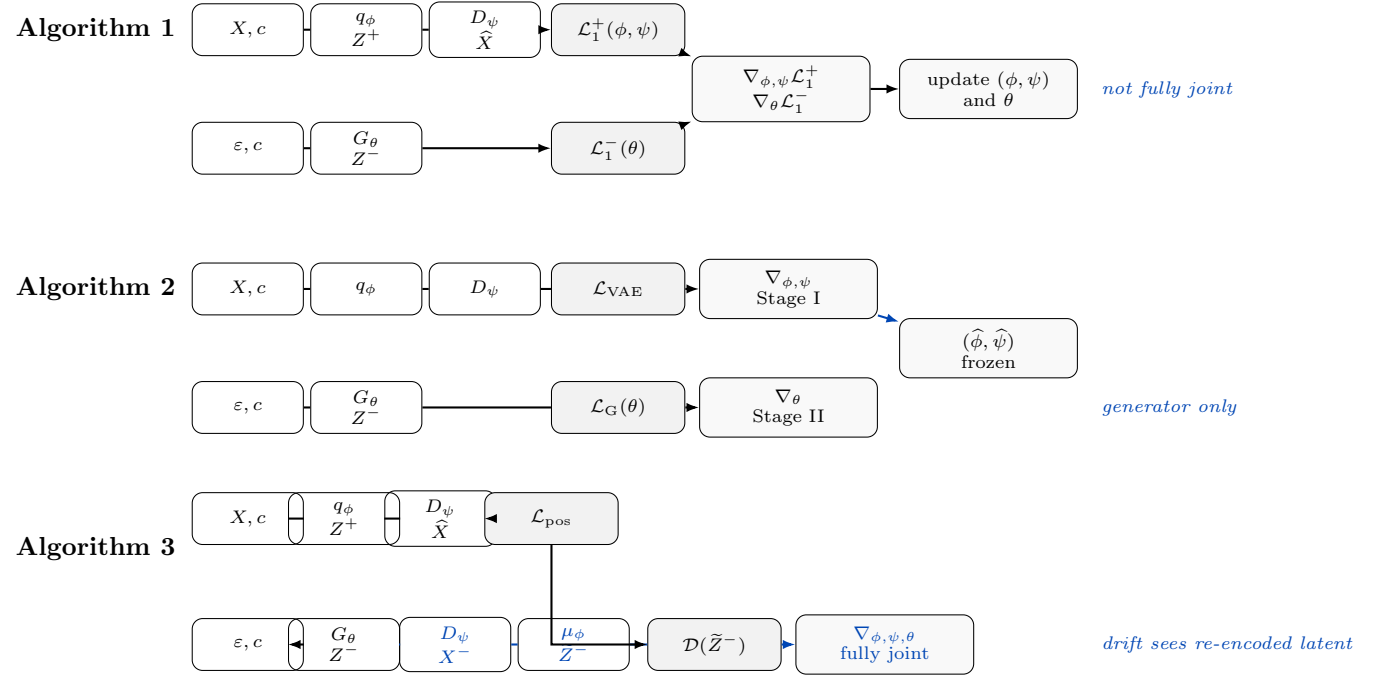


Figure 1: Loss-to-parameter flow. Algorithm 1 has two disconnected gradient blocks despite using one optimizer loop; Algorithm 2 separates VAE and generator stages; Algorithm 3 connects the generated latent to all three parameter blocks through decode-re-encode. Stop-gradient targets do not block gradients through the differentiable first arguments.